

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Miniature cylindrical ion trap mass spectrometry

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ABSTRACT

The development of miniature ion trap mass spectrometers is being driven by many new applications, such as, autonomous underwater vehicles, planetary probes, monitoring air quality in spacecraft and field portable mass spectrometers for the detection of BW/CW agents. Miniaturization of the trap electrodes to sub-millimeter sizes significantly reduces the working voltages and pumping requirements, resulting in a substantial reduction in size and power consumption compared to a conventional instrument, without a significant degradation in overall performance. The cylindrical traps used in the present work are also more easily constructed and are more compatible with micro fabrication techniques than conventional hyperbolic traps. The higher order field components present in such devices can also result in improved performance. The stability diagrams of cylindrical traps with $r_0 = 0.5$ mm have been studied when an axial modulation is applied to the end caps of the trap. Ion ejection from normally stable trajectories was observed when the ion secular frequency was approximately that of a nonlinear resonance of the trap. The resonances are attributed to electrical as well as geometrical considerations. Under these conditions, both the peak height and the mass resolution of the resulting mass spectrum are found to increase.